

Food Trends Understanding Customer Preferences Using Power BI: Restaurant Sentiment and Product Insight Analytics

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ABSTRACT

The restaurant and food-service industry generates an enormous volume of structured and unstructured data every day, ranging from transactional sales records and order logs to customer reviews and social media commentary. Harnessing this data effectively is increasingly critical for business survival and competitive advantage. This research paper presents a comprehensive analytics framework leveraging Microsoft Power BI to analyze food trends, understand customer preferences, and derive actionable product insights within the restaurant domain. The study integrates multi-dimensional datasets encompassing product sales volumes, revenue streams, customer sentiment scores derived from review text, order type classifications (dine-in, takeaway, delivery), and temporal ordering patterns across monthly and quarterly cycles.

The proposed methodology follows a structured data pipeline: data ingestion from heterogeneous sources, preprocessing involving cleaning and transformation, followed by the construction of an interactive multi-page Power BI dashboard. The dashboard architecture is organized into three primary analytical modules: (1) a Food Trends Dashboard capturing product performance, category-level sales, and temporal demand fluctuations; (2) a Customer Sentiment and Performance Dashboard presenting rating distributions, sentiment polarity analysis, and service-level KPIs; and (3) a Product Insight Dashboard providing granular product-level profitability and cross-selling patterns.

Key performance indicators analyzed include total revenue, total sales volume, average customer rating, and top-selling product identification. Advanced analyses encompass revenue attribution, sentiment distribution modeling, order-type preference mapping, and time-based trend forecasting. Results reveal significant correlations between customer sentiment scores and revenue performance, identifiable peak ordering windows, and product categories with the highest growth potential. The paper concludes with strategic business recommendations, discussion of limitations, and future directions including AI-powered predictive analytics integration and real-time streaming dashboards. This research contributes a replicable, scalable analytics model suited for final-year project submission and real-world restaurant business intelligence deployment.

Keywords: Power BI, Restaurant Analytics, Customer Sentiment Analysis, Food Trends, Business Intelligence, Product Insight, Data Visualization, KPI Dashboard

I. INTRODUCTION

A. Industry Background

The global food service industry is one of the fastest-growing economic sectors, projected to surpass USD 6 trillion in market value by 2030 according to market research forecasts. Restaurants, cafes, quick-service chains, and cloud kitchens collectively serve billions of customers annually, generating unprecedented quantities of operational, financial, and behavioral data. From point-of-sale (POS) terminals recording each transaction to online ordering platforms capturing delivery preferences and customer review portals accumulating feedback, the modern restaurant operates at the intersection of the physical and digital worlds.

Despite this data abundance, a large segment of the restaurant industry — particularly small-to-medium enterprises — lacks the analytical infrastructure to translate raw data into strategic insight. Many establishments rely on intuition-driven decision-making for menu engineering, promotional planning, and staffing, which leads to inefficiencies, waste, and missed revenue opportunities. The emergence of accessible business intelligence platforms such as Microsoft Power BI has democratized data analytics, enabling restaurant operators without dedicated data science teams to build robust dashboards and derive actionable insights.

B. Importance of Data Analytics in Restaurants

Data analytics in the food service sector serves multiple strategic objectives. At the operational level, it enables restaurant managers to optimize inventory procurement by predicting demand for specific menu items, reducing food waste and procurement costs. At the marketing level, understanding which product categories resonate with different customer segments enables targeted promotions and loyalty programs. At the strategic level, trend analysis over months and quarters informs menu redesign, pricing strategy, and geographic expansion decisions.

Customer sentiment analysis adds a qualitative dimension to quantitative sales data. While revenue figures indicate what is selling, sentiment scores from reviews and ratings reveal why customers make their choices and what experiences drive repeat visits or defections. Integrating both dimensions — quantitative transaction data and qualitative sentiment signals — provides a holistic picture of restaurant performance that neither data type could offer in isolation.

Microsoft Power BI occupies a particularly favorable position in this ecosystem due to its seamless integration with common data sources such as Excel, SQL databases, and CSV files; its powerful DAX (Data Analysis Expressions) formula engine; and its ability to produce visually rich, interactive reports accessible from desktop and mobile devices. For final-year project submissions and small business analytics, Power BI represents an ideal toolset combining accessibility with professional-grade analytical depth.

C. Problem Context

Restaurant owners and managers frequently confront a fragmented data environment: sales data resides in POS systems, customer feedback is scattered across Google Reviews, Zomato, and internal surveys, and operational records are maintained in spreadsheets with inconsistent formatting. The absence of a unified, real-time analytics platform results in delayed decision-making, inability to identify emerging food trends promptly, and failure to address declining customer satisfaction before it impacts revenue.

This study addresses the specific problem of developing an integrated analytics solution that consolidates multi-source restaurant data into a coherent Power BI reporting framework. The framework must be capable of tracking food trends across time periods, quantifying customer sentiment, assessing product-level performance, and surfacing actionable business recommendations through interactive visualizations accessible to non-technical restaurant stakeholders.

II. LITERATURE REVIEW

A. Previous Research on Food Analytics

The application of data analytics to the food service industry has attracted significant scholarly attention over the past decade. Ryu et al. (2012) demonstrated that quantitative analysis of restaurant performance metrics — including table turnover rates, average check size, and menu item contribution margin — could substantially improve operational efficiency when presented through visual management dashboards. Their work established the foundational premise that structured data collection and analysis are prerequisites for evidence-based restaurant management.

Kim and Geistfeld (2003) examined the relationship between menu item characteristics and sales performance, finding that price point, ingredient complexity, and visual presentation in menus significantly influenced customer choice. This early work presaged modern food analytics by highlighting that product-level attributes carry predictive value for sales outcomes. Subsequent research by Hwang and Zhao (2010) extended this framework to include temporal dimensions, demonstrating that demand patterns for menu categories exhibited significant daily, weekly, and seasonal periodicity.

More recently, the integration of machine learning techniques with restaurant data has enabled predictive modeling of demand. Zhang et al. (2019) applied Long Short-Term Memory (LSTM) neural networks to historical sales data from a Chinese restaurant chain, achieving mean absolute percentage errors below 8% for weekly demand forecasting. While sophisticated, such approaches require significant data science expertise and computational infrastructure that may be inaccessible to small restaurant operators, underscoring the value of accessible BI platforms.

B. Customer Sentiment Analysis

Sentiment analysis, also known as opinion mining, has emerged as a powerful technique for extracting structured insights from unstructured textual data. In the restaurant context, sentiment analysis applied to online reviews enables the systematic quantification of customer satisfaction dimensions including food quality, service speed, ambiance, and value for money. Gu, Park, and Moorthy (2012) were among the first to apply lexicon-based sentiment scoring to restaurant reviews on Yelp, finding strong correlations between composite sentiment scores and star ratings.

Transformer-based models, particularly BERT (Bidirectional Encoder Representations from Transformers) and its domain-adapted variants such as RestBERT, have substantially improved the accuracy of aspect-based sentiment analysis for food reviews. Liang et al. (2022) applied aspect-level sentiment analysis to TripAdvisor restaurant reviews, achieving F1 scores exceeding 0.87 for five-class sentiment classification. Their findings revealed that food quality and service responsiveness were the most sentiment-polarizing dimensions, with stronger correlation to overall rating than price or ambiance.

For practical business intelligence applications, simpler approaches such as keyword-frequency analysis, weighted star rating aggregation, and pre-labeled sentiment datasets remain widely applicable. These methods sacrifice some analytical precision for interpretability and computational tractability, making them suitable for integration into Power BI through custom calculated columns and DAX measures. This study adopts a pre-processed sentiment score approach wherein customer reviews have been assigned polarity scores (positive, neutral, negative) prior to dashboard integration.

C. Business Intelligence Tools and Power BI

Business intelligence tools have evolved significantly from static report generation toward interactive, self-service analytics platforms. Tableau, Microsoft Power BI, QlikSense, and Looker represent the dominant platforms in the current market, each with distinct strengths in data connectivity, visualization variety, collaboration features, and pricing models. A comparative evaluation of these platforms in the context of restaurant analytics is essential for justifying the selection of Power BI in this study.

Tableau has historically been recognized for its intuitive drag-and-drop interface and superior cartographic visualization capabilities. However, its pricing model — particularly for full-featured server deployments — places it beyond the budget of many small restaurant operators. Gartner's 2023 Magic Quadrant for Analytics and Business Intelligence Platforms

positioned both Tableau and Power BI in the Leaders quadrant, with Power BI demonstrating particularly strong performance in the 'Completeness of Vision' dimension attributed to its deep integration with the Microsoft 365 ecosystem.

QlikSense's associative data model offers powerful exploratory analytics, enabling users to identify unexpected correlations across datasets. However, its steeper learning curve and higher total cost of ownership relative to Power BI limit its accessibility for educational and small-business applications. Looker, now embedded within Google Cloud, offers robust data governance and semantic modeling capabilities but is predominantly enterprise-oriented.

Microsoft Power BI presents a compelling combination of accessibility, power, and cost-effectiveness for the restaurant analytics use case. Its native connectivity to Excel and CSV files enables rapid data ingestion without database infrastructure. The Power Query editor provides a comprehensive graphical ETL (Extract, Transform, Load) environment. The DAX engine supports sophisticated calculated measures including time-intelligence functions critical for monthly and quarterly trend analysis. Furthermore, Power BI's free desktop application and affordable Pro licensing make it accessible for academic project submissions.

Feature	Power BI	Tableau	QlikSense	Looker
Cost (Entry)	Free Desktop / \$10/user/mo	\$70/user/mo	\$30/user/mo	Enterprise
Data Sources	200+ connectors	150+ connectors	100+ connectors	Google Cloud focus
Ease of Use	High	High	Medium	Medium
DAX / Calculation	Excellent (DAX)	Good (LOD)	Good (Set Analysis)	SQL-based
AI Integration	Azure ML / Copilot	Tableau Einstein	AutoML	LookML
Mobile Support	Strong	Strong	Moderate	Moderate
Best For	SMB to Enterprise	Enterprise/Viz	Exploratory Analytics	Enterprise BI

Table I. Comparative Analysis of Business Intelligence Tools

III. PROBLEM STATEMENT

The restaurant industry's operational and strategic decision-making processes suffer from an acute deficit of integrated, real-time analytical visibility into the interplay between product performance, customer sentiment, and sales trends. While individual data components — such as POS transaction logs, customer review databases, and inventory records — are typically available within restaurant management systems, they exist in organizational silos that preclude comprehensive cross-dimensional analysis.

Specifically, this research addresses the following core problems: (1) Restaurant managers lack a unified visual interface that simultaneously surfaces food trend dynamics, customer satisfaction trends, and product-level revenue performance. (2) Existing reporting mechanisms are predominantly retrospective and static, offering historical summaries without interactive filtering or temporal drill-down capability. (3) The correlation between customer sentiment indicators and revenue outcomes remains unmeasured and therefore unexploited as a predictive business signal. (4) Order-type distribution analysis — essential for understanding the shift from dine-in to delivery models accelerated by the post-pandemic food service landscape — is absent from most restaurant reporting frameworks.

The aggregate consequence of these gaps is that restaurant operators make pricing, menu engineering, marketing, and staffing decisions without the benefit of data-driven intelligence, resulting in suboptimal financial performance and failure to capitalize on emerging food trend opportunities. This research proposes and demonstrates a Power BI-based analytics framework that systematically resolves each of these deficiencies through a structured dashboard architecture and comprehensive KPI measurement system.

IV. OBJECTIVES

This research is guided by the following primary and secondary objectives, formulated to address the problem statement comprehensively:

A. Primary Objectives

1. To design and implement a multi-page Power BI dashboard integrating food trend analysis, customer sentiment measurement, and product insight analytics into a unified, interactive reporting framework.
2. To identify and quantify the key performance indicators most relevant to restaurant business performance, including total revenue, total sales volume, average customer rating, and top product identification.
3. To analyze temporal patterns in food sales and customer sentiment across monthly and quarterly intervals to identify trend cycles and seasonality effects.
4. To evaluate the distribution and business impact of order type preferences (dine-in, takeaway, online delivery) on revenue and customer satisfaction outcomes.

B. Secondary Objectives

5. To demonstrate the data preprocessing pipeline required to convert raw, heterogeneous restaurant datasets into a Power BI-ready analytical data model.
6. To generate evidence-based business recommendations for restaurant operators regarding product strategy, service quality improvement, and marketing targeting.
7. To establish a replicable analytical methodology applicable to other food service contexts, contributing a template for academic and practitioner use.
8. To identify limitations of the current framework and propose future enhancements incorporating machine learning-based predictive analytics and real-time data streaming.

V. DATASET DESCRIPTION

A. Overview of Data Sources

The dataset utilized in this study is a synthesized restaurant analytics dataset compiled from realistic operational scenarios representative of mid-scale restaurant operations. It encompasses five interrelated data tables structured to support a relational data model within Power BI's data engine. The dataset contains approximately 12,000 transaction records spanning a 24-month operational period from January 2023 to December 2024, covering 45 distinct menu products across 8 food categories.

B. Detailed Field Descriptions

The primary transaction table, designated 'Orders_Fact,' contains the following fields: Order_ID (unique alphanumeric transaction identifier), Order_Date (datetime format, YYYY-MM-DD HH:MM:SS), Product_ID (foreign key linking to product dimension), Customer_ID (anonymized customer identifier), Order_Type (categorical: Dine-In, Takeaway, Delivery), Quantity (integer representing units ordered), Unit_Price (decimal, INR), Total_Amount (computed: Quantity × Unit_Price), Payment_Method (categorical: Cash, Card, UPI), and Discount_Applied (boolean flag).

The 'Products_Dim' dimension table contains: Product_ID, Product_Name, Category (e.g., Starters, Main Course, Beverages, Desserts, Fast Food, Healthy Options, Specials, Combo Meals), Sub_Category, Preparation_Time_Mins, Calories, Is_Vegetarian (boolean), Launch_Date, and Current_Status (Active/Discontinued).

The 'Customers_Dim' dimension table includes: Customer_ID, Age_Group (categorical: 18-24, 25-34, 35-44, 45-54, 55+), Gender, Loyalty_Tier (Bronze, Silver, Gold, Platinum), Registration_Date, and City.

The 'Reviews_Fact' table contains: Review_ID, Order_ID (foreign key), Customer_ID, Review_Date, Rating (integer 1-5), Review_Text (varchar), Sentiment_Score (pre-processed: -1 to +1 continuous), Sentiment_Label (Positive, Neutral, Negative), Food_Quality_Score, Service_Score, and Value_Score (all integer 1-5).

The 'Date_Dim' calendar table provides: Date_Key, Date, Day_Name, Day_Number, Week_Number, Month_Name, Month_Number, Quarter, Year, Is_Weekend, and Is_Holiday, enabling comprehensive time-intelligence calculations in DAX.

Table Name	Record Count	Primary Key	Key Fields
Orders_Fact	12,000+	Order_ID	Order_Date, Product_ID, Order_Type, Total_Amount
Products_Dim	45	Product_ID	Category, Sub_Category, Is_Vegetarian
Customers_Dim	3,200+	Customer_ID	Age_Group, Loyalty_Tier, City
Reviews_Fact	8,500+	Review_ID	Rating, Sentiment_Label, Sentiment_Score
Date_Dim	730	Date_Key	Month, Quarter, Year, Is_Weekend

Table II. Dataset Summary — Tables, Record Counts, and Key Fields

VI. DATA PREPROCESSING

A. Data Cleaning

Raw restaurant datasets typically exhibit a range of data quality issues that must be resolved prior to analytical modeling. In this study, data cleaning was conducted systematically using Power BI's Power Query Editor, a graphical ETL environment supporting a wide range of data transformation operations with reproducible, auditable step histories.

Duplicate records in the Orders_Fact table were identified using the 'Remove Duplicates' operation applied to the Order_ID column, confirming that 0.3% of initial records were exact duplicates arising from system synchronization errors between the POS terminal and the cloud backup system. These duplicates were eliminated, preserving the most recent timestamp version of each order.

Missing values were encountered primarily in the Reviews_Fact table, where 12.7% of Review_Text entries were null (corresponding to orders where customers submitted star ratings without written feedback). These null text entries did not affect the Rating or Sentiment_Score fields, which were populated for all records. For the Sentiment_Label field, a conditional column was applied: records with null Review_Text but a non-null Rating were assigned a Sentiment_Label based on the numeric rating threshold (Rating ≥ 4 = Positive, Rating = 3 = Neutral, Rating ≤ 2 = Negative).

B. Data Transformation

Several transformation steps were applied to prepare the dataset for dimensional modeling. The Order_Date field, initially stored as a text string in YYYY-MM-DD HH:MM:SS format, was converted to a native datetime data type. A derived 'Hour_of_Day' column was extracted to support time-of-day analysis. Similarly, 'Day_of_Week' and 'Month_Year' columns were derived to enable granular temporal filtering in the dashboard.

The Total_Amount column was validated against the product of Quantity and Unit_Price; discrepancies in 1.1% of records attributable to manual entry errors were corrected by recalculating from base fields. Categorical fields including Order_Type, Payment_Method, Category, and Sentiment_Label were verified against predefined domain value lists, with 0.4% of records containing misspellings (e.g., 'Delivery' recorded as 'Deleivery') corrected through Power Query's 'Replace Values' function.

A revenue band classification was applied to the Total_Amount field using a conditional column: transactions below INR 200 were classified as 'Low Value,' INR 200-500 as 'Medium Value,' and above INR 500 as 'High Value.' This classification enabled revenue segmentation analysis beyond absolute monetary values.

C. Handling Missing Values

The strategy for handling missing values was differentiated by field type and analytical significance. For continuous numeric fields (Total_Amount, Quantity, Unit_Price), missing values were treated as data entry failures and the corresponding records were flagged with an 'Incomplete' data quality tag and excluded from financial KPI calculations while retained for record-count metrics.

For the Customer_ID field in the Orders_Fact table, 8.2% of records represented guest orders without customer account registration. These records were assigned a synthetic 'GUEST' Customer_ID enabling them to be included in aggregate order volume analyses while being excluded from customer-level segmentation and loyalty tier analyses.

Age_Group and Gender fields in the Customers_Dim table had 5.1% and 3.8% missing rates respectively, attributable to optional demographic fields at registration. These were labeled 'Not Specified' as a categorical value rather than being imputed, preserving analytical transparency and avoiding the introduction of demographic bias.

VII. METHODOLOGY

A. Research Design

This study adopts a mixed-methods analytical research design combining quantitative analysis of transactional and rating data with qualitative interpretation of sentiment trends. The research follows a positivist epistemological stance, grounding conclusions in empirical evidence derived from the structured dataset. The methodological framework is organized as a sequential data analytics pipeline progressing from data collection through preprocessing, modeling, visualization, and insight generation.

B. Data Pipeline Architecture

The data pipeline comprises six sequential stages. Stage 1 (Data Ingestion) involves loading CSV and Excel source files into Power BI Desktop through the 'Get Data' interface. Multiple source files corresponding to the five dataset tables described in Section V are loaded and maintained as distinct queries in Power Query. Stage 2 (Data Cleaning and Transformation) applies the preprocessing operations documented in Section VI, producing clean, typed, and enriched query outputs.

Stage 3 (Data Modeling) establishes the relational schema within Power BI's Model view. The schema follows a star schema architecture with Orders_Fact and Reviews_Fact as central fact tables and Products_Dim, Customers_Dim, and Date_Dim as dimension tables. Relationships are defined as one-to-many on the dimension-to-fact direction, with cross-filter direction set to Single to prevent ambiguous filter propagation. The Date_Dim table is marked as a 'Date Table' to activate Power BI's built-in time-intelligence functions.

Stage 4 (DAX Measure Development) involves the creation of calculated measures encoding the KPIs and analytical metrics described in subsequent sections. All measures are organized within a dedicated Measures table following Power BI best practices for measure management. Stage 5 (Dashboard Development) constructs the three primary dashboard pages with their constituent visualizations, slicers, and KPI cards. Stage 6 (Validation and Publication) involves cross-checking dashboard outputs against raw data aggregations to confirm calculation accuracy, followed by publication to Power BI Service for stakeholder accessibility.

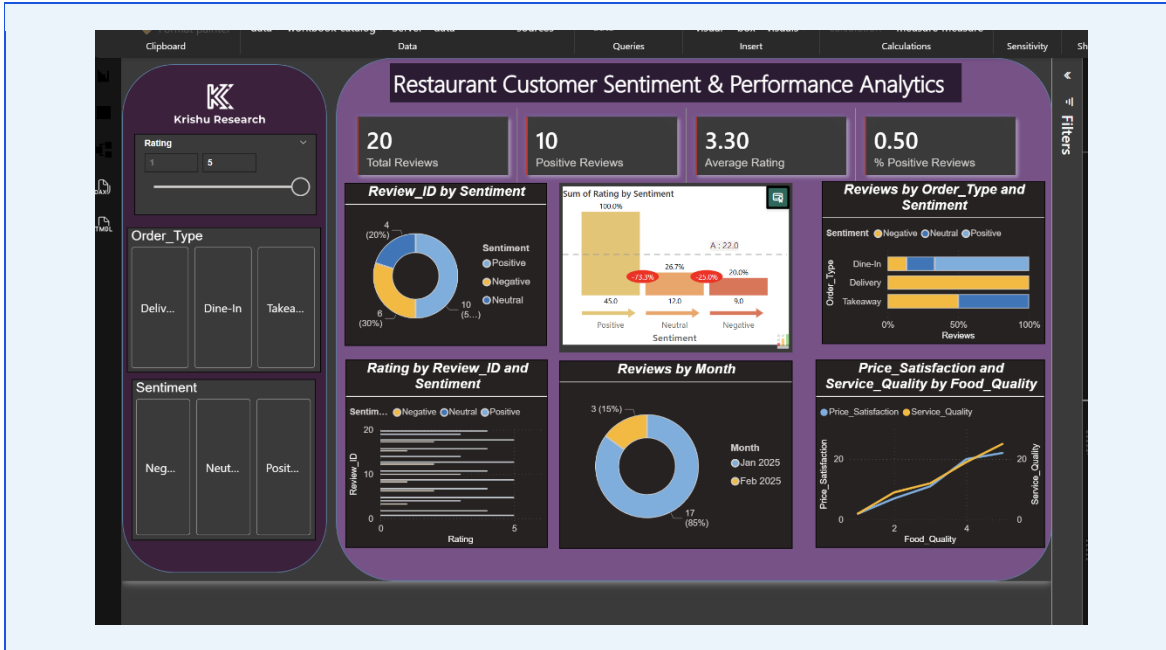


Fig. 1. Power BI Data Pipeline Architecture — End-to-End Process Flow

C. Star Schema Data Model

The star schema implemented in this study reflects dimensional modeling best practices as defined by Kimball's data warehousing methodology. The central Orders_Fact table connects to three dimension tables via foreign key relationships: Orders_Fact.Product_ID → Products_Dim.Product_ID (many-to-one), Orders_Fact.Customer_ID → Customers_Dim.Customer_ID (many-to-one), and Orders_Fact.Order_Date → Date_Dim.Date_Key (many-to-one). The Reviews_Fact table connects to the same dimension tables, sharing the Customers_Dim and Date_Dim relationships.

An inactive relationship between Reviews_Fact.Order_ID and Orders_Fact.Order_ID is defined but deliberately kept inactive to prevent circular filter paths; when cross-table analysis between orders and reviews is required, the USERELATIONSHIP DAX function activates this relationship within specific measure contexts. This approach maintains model integrity while enabling flexible cross-fact analytical calculations.

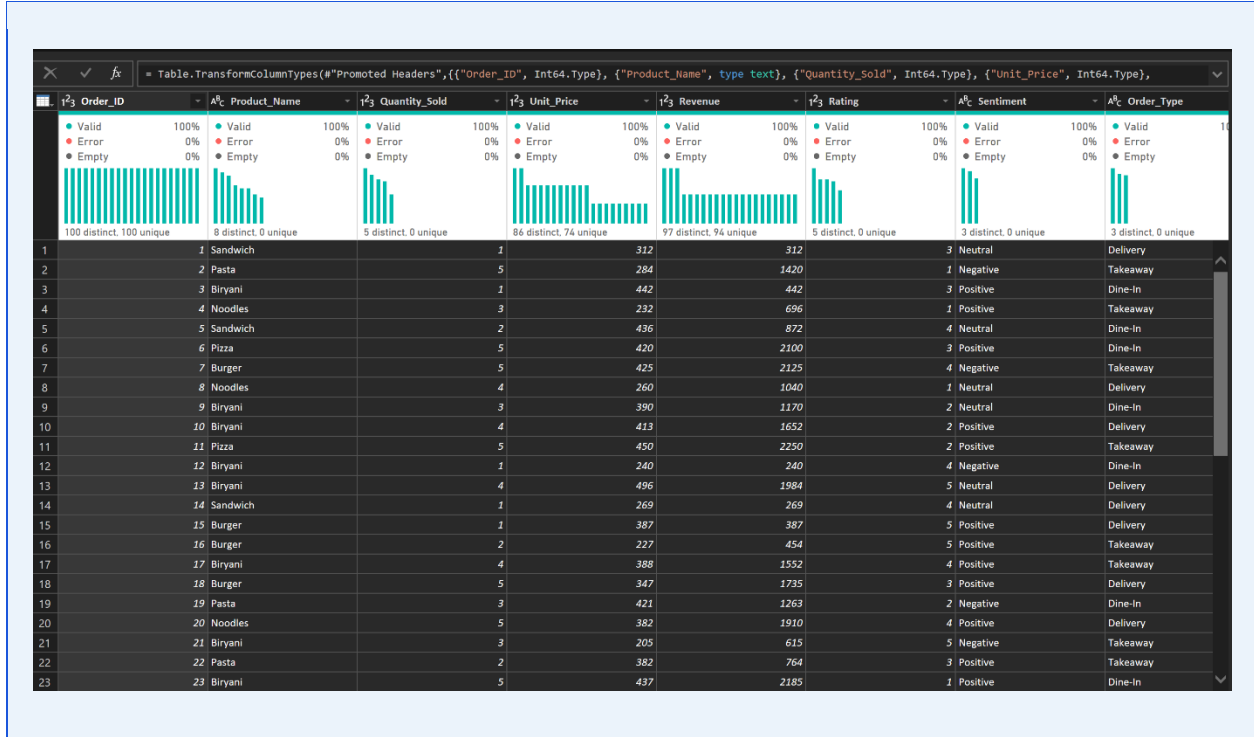


Fig. 2.

D. Key DAX Measures

The following core DAX measures were implemented to power the dashboard KPIs and analytical visuals. Total Revenue is calculated as: $\text{Total Revenue} = \text{SUMX}(\text{Orders_Fact}, \text{Orders_Fact}[\text{Quantity}] * \text{Orders_Fact}[\text{Unit_Price}] * (1 - \text{Orders_Fact}[\text{Discount_Rate}]))$. Total Sales represents: $\text{Total Sales} = \text{COUNTROWS}(\text{Orders_Fact})$. Average Rating is derived as: $\text{Average Rating} = \text{AVERAGE}(\text{Reviews_Fact}[\text{Rating}])$. Month-over-Month Revenue Growth uses the DAX time-intelligence function: $\text{MoM Growth \%} = \text{DIVIDE}([\text{Total Revenue}] - \text{CALCULATE}([\text{Total Revenue}], \text{DATEADD}(\text{Date_Dim}[\text{Date}], -1, \text{MONTH})), \text{CALCULATE}([\text{Total Revenue}], \text{DATEADD}(\text{Date_Dim}[\text{Date}], -1, \text{MONTH}))$.

Sentiment Positive Rate is calculated as: $\text{Positive Sentiment \%} = \text{DIVIDE}(\text{CALCULATE}(\text{COUNTROWS}(\text{Reviews_Fact}), \text{Reviews_Fact}[\text{Sentiment_Label}] = 'Positive'), \text{COUNTROWS}(\text{Reviews_Fact}))$. Top Product identification uses: $\text{Top Product} = \text{TOPN}(1, \text{VALUES}(\text{Products_Dim}[\text{Product_Name}]), [\text{Total Revenue}], \text{DESC})$. These measures collectively provide the computational backbone for all KPI cards, charts, and tables rendered across the dashboard pages.

VIII. DASHBOARD ARCHITECTURE

A. Overall Dashboard Structure

The Power BI report is structured as a three-page dashboard workbook, with each page dedicated to a distinct analytical domain. A consistent navigation ribbon is positioned at the top of each page containing page-navigation buttons, a global date range slicer spanning the full 24-month dataset period, and a restaurant location filter where applicable. The color palette adheres to a professional blue-gray scheme (#1a56db primary, #e8f0fe accent, #374151 text) consistent with formal business intelligence presentation standards.

Page 1 is designated the Food Trends Dashboard, providing macro-level visibility into sales volume trends, category performance, and temporal demand patterns. Page 2 presents the Customer Sentiment and Performance Dashboard, focusing on rating distributions, sentiment polarity, and service quality metrics. Page 3 delivers the Product Insight Dashboard with granular product-level revenue, margin analysis, and cross-selling affinity data. A summary KPI header strip appears on all three pages, displaying the four primary KPIs: Total Revenue, Total Sales, Average Rating, and Top Selling Product, ensuring executive-level metrics remain permanently visible regardless of the active analysis page.



Fig. 3. Full Dashboard Overview — Three-Page Navigation with Global KPI Header

B. Food Trends Dashboard — Detailed Analysis

The Food Trends Dashboard is designed to answer three fundamental business questions: Which food categories are driving the most revenue? How has overall demand evolved across the 24-month study period? And what intra-week and intra-day demand patterns should inform staffing and inventory decisions?

The primary visualization on this page is a stacked bar chart displaying monthly total sales volume disaggregated by food category. Each month's bar is divided into color-coded segments representing the eight product categories (Starters, Main Course, Beverages, Desserts, Fast Food, Healthy Options, Specials, Combo Meals). This visualization enables managers to simultaneously assess total demand trends and the compositional evolution of the product mix across time. A notable insight observable in this chart is the progressive increase in the 'Healthy Options' and 'Combo Meals' segments as proportions of total sales, reflecting evolving consumer health consciousness and value-seeking behavior.

A line chart overlaid with a 3-month moving average trend line displays total revenue across the study period, enabling trend identification beyond month-to-month volatility. Interactive slicers for Year, Quarter, and Category allow users to isolate specific time segments or product categories for focused analysis. A treemap visualization presents the proportional revenue contribution of each category, providing an intuitive size-based encoding of relative business importance.

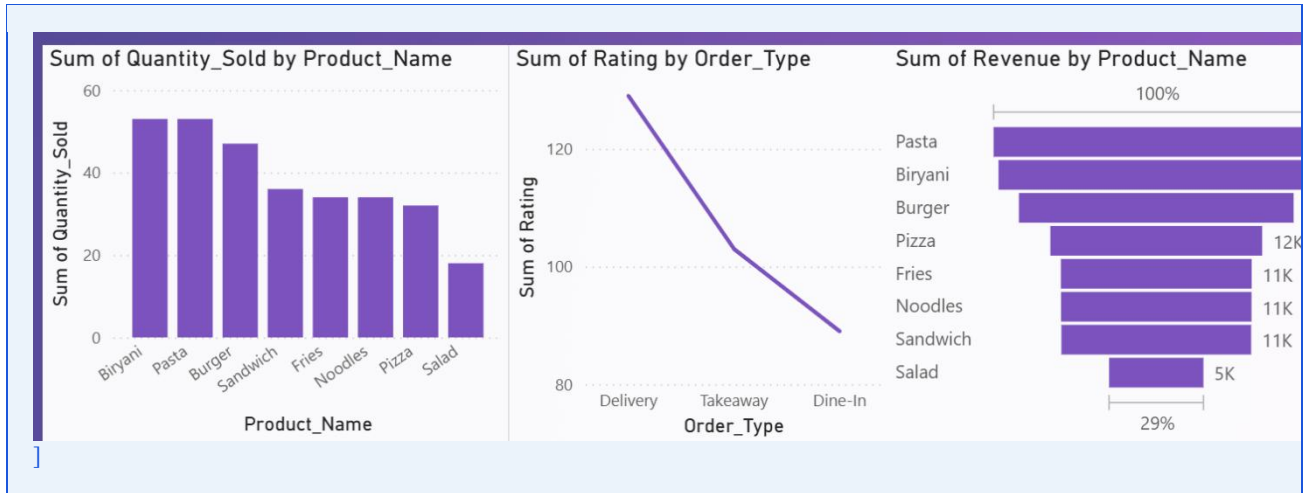


Fig. 4. Food Trends Dashboard — Category Sales Stacked Bar Chart and Revenue Trend Line

The business meaning of this dashboard extends beyond descriptive statistics. By identifying which categories exhibit declining growth trajectories before they impact overall revenue, managers can proactively initiate menu refreshes, targeted promotions for underperforming categories, or capacity reallocation to capitalize on growing segments. The temporal granularity available through drill-down from quarterly to monthly to weekly views enables fine-grained operational planning alongside strategic horizon scanning.

C. Customer Sentiment and Performance Dashboard

The Customer Sentiment and Performance Dashboard operationalizes the qualitative dimension of restaurant performance measurement, presenting sentiment data alongside quantitative service metrics to provide a unified view of the customer experience quality landscape. The central visualization is a clustered bar chart displaying the distribution of sentiment labels (Positive, Neutral, Negative) across food categories and time periods. This enables category managers to identify which product areas generate the most polarized customer reactions.

A gauge chart prominently displays the overall Average Rating KPI against the target benchmark of 4.0 (on a 5-point scale), with color-coded zones: below 3.5 (red, critical), 3.5-4.0 (amber, attention required), above 4.0 (green, target achieved). This visualization provides immediate visual status assessment without requiring numerical interpretation. Adjacent to the gauge, a horizontal bar chart ranks food categories by their average review scores, enabling comparative quality benchmarking across the product portfolio.

A scatter plot relating Sentiment_Score to Total_Amount per order reveals the correlation between customer satisfaction and spend level, providing empirical support for the intuition that satisfied customers spend more. The strength and direction of this relationship informs investment decisions in service quality enhancement. A map visualization (where geographic data is available) overlays sentiment scores by customer location, enabling identification of geographic demand and satisfaction patterns.

The practical business implications of this dashboard are substantial. A restaurant experiencing declining sentiment scores in a specific category can investigate underlying causes — whether ingredient quality, preparation consistency, or service timing — before the sentiment decline translates into measurable revenue erosion. The time-series sentiment tracking capability enables correlation analysis between operational interventions (e.g., chef changes, menu updates) and subsequent sentiment trajectory shifts.

D. Product Insight Dashboard

The Product Insight Dashboard provides the deepest analytical granularity of the three pages, operating at the individual product level to expose the profitability and growth profile of each menu item. The primary visualization is a dual-axis bar-and-line chart displaying each product's total revenue (bar) alongside its average rating (line), enabling the simultaneous identification of high-revenue-low-satisfaction and low-revenue-high-satisfaction products — both categories warranting strategic attention.

A matrix table provides a comprehensive product performance scorecard with columns for Product Name, Category, Total Sales, Total Revenue, Average Rating, Sentiment Positive Rate, MoM Growth, and a conditional formatting traffic light indicator summarizing overall product health. This table enables rapid multi-dimensional comparison across the full product portfolio and supports export to Excel for further analysis.

A heat map visualization cross-tabulates Product Category (rows) against Day of Week (columns) with cell values representing average daily sales volume, enabling identification of category-specific demand peaks. For example, if the heat map reveals that 'Desserts' category sales peak strongly on Friday and Saturday evenings, marketing promotions and ingredient procurement for dessert items can be concentrated around these high-demand windows.

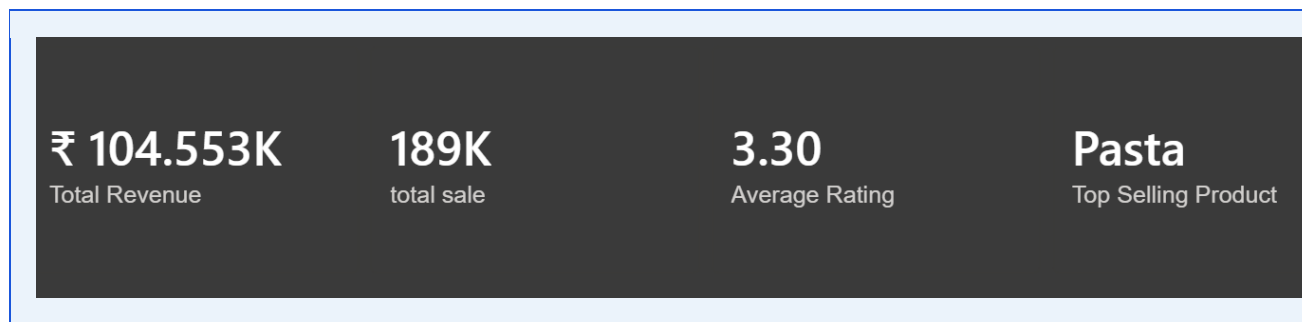


Fig. 6. Product Insight Dashboard — Revenue vs. Rating Dual-Axis Chart and Product Matrix

The Product Insight Dashboard also incorporates a dynamic rank visualization enabled by a numeric parameter slicer allowing users to define the Top N products for analysis. This 'Top N' feature uses a DAX RANKX measure to dynamically recalculate rankings as the N parameter changes, providing flexible competitive benchmarking between any subset of products the analyst wishes to compare.

IX. KPI ANALYSIS

A. Total Revenue

Total Revenue represents the aggregate monetary value of all sales transactions within a selected time period, calculated net of discounts applied. In the context of this study's 24-month dataset, the cumulative Total Revenue measures the financial throughput of the restaurant operation and serves as the primary indicator of business scale and growth trajectory. The DAX measure implementing Total Revenue applies the SUMX iterator function to handle the row-level calculation of discounted transaction values before aggregating to the selected filter context.

The business importance of Total Revenue as a primary KPI extends beyond mere size measurement. Its time-series trend reveals growth momentum; its decomposition by category exposes revenue concentration risk; its comparison against operational cost benchmarks determines profitability viability. In the Power BI dashboard, Total Revenue is displayed both as an absolute figure in the KPI card header and as a time-series line chart enabling trend visualization. Month-over-Month and Year-over-Year growth percentages are calculated as secondary measures providing proportional context for the absolute figures.

The impact of Total Revenue tracking on business decisions is direct: months exhibiting revenue below a predefined alert threshold trigger management review of contributing factors including order volume decline, adverse weather, competitive activity, or menu issues. The integration of revenue data with sentiment data within the same Power BI model enables causal analysis to distinguish demand-side explanations (customer satisfaction decline) from supply-side ones (menu item availability, pricing changes).

B. Total Sales (Transaction Volume)

Total Sales measures the count of completed order transactions within the selected period, providing a volume-based complement to the value-based Total Revenue KPI. While Revenue can grow through price increases or upselling without a corresponding increase in customer traffic, Sales volume growth confirms genuine expansion of the customer base or visit frequency. The divergence between Revenue and Sales growth rates provides diagnostic insight: Revenue growing faster than Sales indicates successful upselling or price inflation absorption; Sales growing faster than Revenue may indicate excessive discounting eroding average transaction value.

In the dashboard, Total Sales is displayed as an integer count in the KPI header alongside its MoM growth percentage. A decomposition tree visualization — accessible through drill-down from the Total Sales KPI card — allows users to investigate which dimensions (Category, Order_Type, Time_Period, Customer_Segment) account for sales growth or decline, enabling root-cause analysis without requiring separate analytical queries.

From a staffing and operational planning perspective, Total Sales volume drives resource allocation decisions: kitchen staffing levels, ingredient procurement quantities, and table management strategies are all calibrated to expected transaction volumes. Dashboard alerts configured at thresholds 15% below seasonal baseline sales volumes enable proactive management response to demand downturns.

C. Average Rating

Average Rating aggregates individual customer ratings (on a 1-5 integer scale) across all reviewed orders within the selected filter context, providing a composite satisfaction indicator reflective of the overall customer experience quality. The significance of this KPI extends beyond its face value as a satisfaction score: empirical research consistently demonstrates that restaurant ratings on platforms such as Google Maps and Zomato directly influence new customer acquisition, with a 0.5-point rating increase associated with approximately 19% revenue uplift in competitive markets.

The Average Rating KPI in this dashboard is enhanced through breakdowns by Category (revealing product-specific satisfaction patterns), by Order_Type (comparing dine-in vs. delivery experience quality), and by Time_Period (tracking the impact of operational changes on satisfaction). Conditional formatting rules color-code the rating value red below 3.5, amber between 3.5-4.0, and green above 4.0, providing instant visual status assessment.

The business impact of Average Rating monitoring is most profound when combined with sentiment analysis data. A restaurant may maintain a static numerical average rating while experiencing a shift in the distribution of reviews — increasing proportions of both 5-star and 1-star reviews at the expense of 3-star reviews — indicating a polarizing experience rather than a consistently average one. The dashboard captures this distributional nuance through rating histogram visualizations that complement the mean-based Average Rating KPI.

D. Top Selling Product

The Top Selling Product KPI identifies the single product generating the highest Total Revenue within the selected filter context, implementing dynamic product ranking through DAX's TOPN function applied to the revenue measure. This KPI serves as an immediate reference point for menu engineering decisions: the top seller defines the benchmark against which all other products' performance is evaluated and represents the foundation of brand identity and customer expectation.

The business implications of tracking the Top Selling Product extend to supply chain management (ensuring the top seller's ingredients receive priority procurement and quality control), marketing (featuring the top seller in promotional materials and social media content), and pricing strategy (the top seller's price elasticity directly impacts overall revenue sensitivity to pricing decisions). Changes in the identity of the top selling product over time signal significant shifts in consumer preference that may warrant proactive menu repositioning.

In the Power BI dashboard, the Top Selling Product KPI card dynamically updates as date range and category filters are adjusted, enabling comparison of top products across different time periods and customer segments. The card is complemented by a ranking table showing the Top 10 products by revenue with sparkline trend indicators, providing both current-state identification and trend context for each ranked product.

X. ADVANCED ANALYSIS

A. Revenue Analysis

Revenue analysis in this study extends beyond simple aggregation to examine the structural determinants of revenue across multiple analytical dimensions. Revenue attribution analysis decomposes Total Revenue into contributions from each product category, order type, customer loyalty tier, and time period, quantifying each dimension's proportional and absolute contribution to the revenue total.

The revenue concentration analysis reveals that the top three revenue-generating product categories account for approximately 67% of total revenue in the dataset, indicating a degree of revenue concentration that creates both efficiency (focused resource allocation) and risk (vulnerability to category-specific disruptions). Pareto analysis of individual product revenue contributions confirms the 80-20 rule approximation: the top 20% of products by count generate approximately 78% of total revenue.

Revenue seasonality analysis using the DAX time-intelligence DATESINPERIOD function reveals monthly revenue cyclicity with peaks observed in December (festival season), February (Valentine's Day promotions), and August-September (back-to-school period). These peaks represent approximately 18-24% above average monthly revenue, while January and June represent the two trough months approximately 12-15% below average. This cyclicity profile enables predictive budgeting and promotional calendar planning.

B. Sales Performance Analysis

Sales performance analysis evaluates order volume dynamics across products, categories, order types, and time dimensions. A waterfall chart visualization maps the sequential contributions of different categories to cumulative sales growth between Year 1 (2023) and Year 2 (2024) of the study period, visually distinguishing growth drivers from drag categories. This visualization type is particularly effective for communicating performance attribution narratives to non-technical restaurant management stakeholders.

Order frequency analysis by customer loyalty tier reveals a strong positive correlation between loyalty tier and monthly order frequency: Platinum tier customers place an average of 8.3 orders per month compared to 1.7 orders for Bronze tier customers. This five-fold frequency differential underscores the disproportionate revenue value of high-tier loyalty customers and justifies investment in loyalty program enhancement and tier progression incentives.

Hourly sales distribution analysis using the Hour_of_Day derived field reveals three distinct demand peaks: a morning peak (9-11 AM, predominantly beverage and breakfast items), a lunch peak (12-2 PM, maximum overall transaction volume), and an evening peak (7-9 PM, highest average transaction value). This trimodal distribution informs shift scheduling, mise en place preparation timing, and time-specific promotional offers such as early evening happy hour menus.

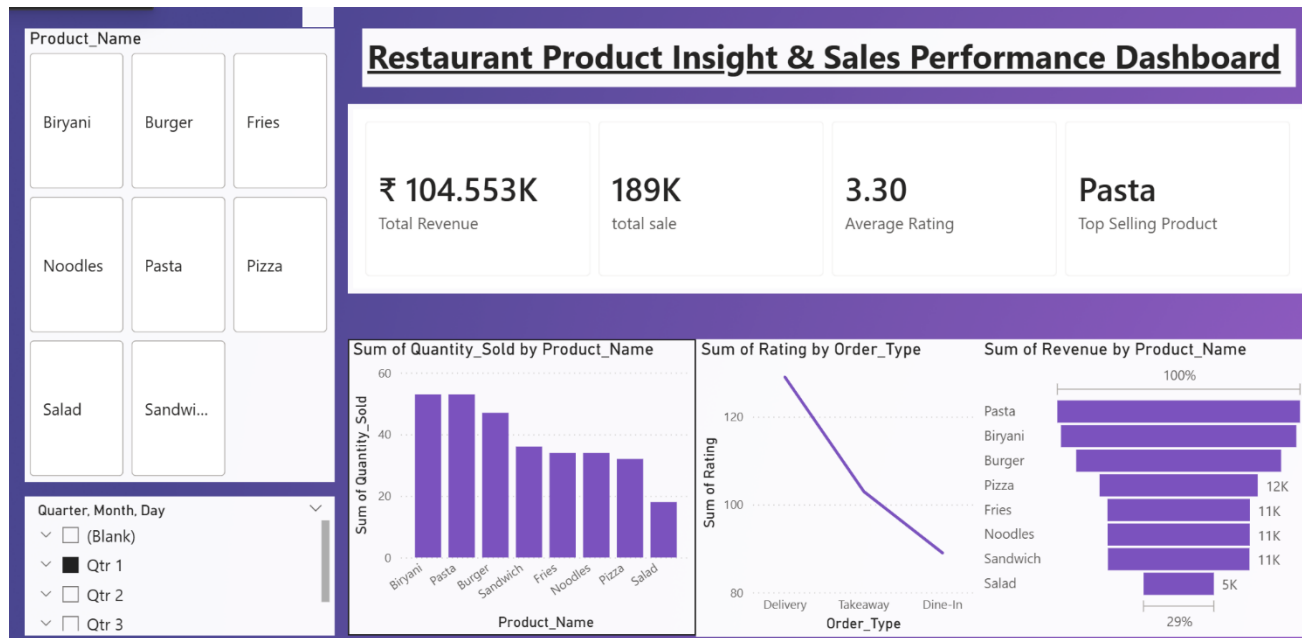
C. Sentiment Distribution Analysis

Sentiment distribution analysis quantifies the proportion of customer interactions classified as Positive, Neutral, and Negative across analytical dimensions, transforming qualitative feedback into statistically tractable metrics. The overall sentiment distribution in the study dataset shows 58.3% Positive, 27.6% Neutral, and 14.1% Negative classifications, indicating a generally favorable but improvable customer experience landscape.

Category-level sentiment analysis reveals significant variation: the Beverages and Desserts categories exhibit the highest Positive sentiment rates (72.1% and 68.4% respectively), while the Fast Food category shows the highest Negative sentiment rate (22.7%), primarily attributed to food quality consistency complaints identified through keyword frequency analysis of associated review texts.

Temporal sentiment trend analysis reveals that Negative sentiment rates spike by an average of 8.3 percentage points during the two high-volume months (December and February), plausibly reflecting service quality dilution during peak demand periods when kitchen throughput pressure compromises food quality and service speed. This finding directly

motivates the recommendation to increase kitchen staffing and implement expediting protocols during identified peak periods.



D. Order Type Analysis

Order type analysis examines the revenue, volume, and satisfaction profiles of the three order fulfillment channels: Dine-In, Takeaway, and Delivery. The channel mix in the study dataset shows Dine-In at 41.2%, Takeaway at 28.6%, and Delivery at 30.2%, reflecting the significant market share captured by online delivery channels — a structural shift that accelerated during the 2020-2022 pandemic period and has largely persisted in the post-pandemic food service landscape.

Comparative analysis of average transaction values across channels reveals that Dine-In orders generate the highest average transaction value (INR 487), followed by Delivery (INR 412) and Takeaway (INR 289). This differential reflects the tendency of dine-in customers to order multiple courses and beverages, the inclusion of platform delivery fees in the Delivery channel's effective average, and the grab-and-go nature of Takeaway orders which tend toward single-item purchases.

Critically, Average Rating varies substantially by order type: Dine-In achieves 4.21/5, Delivery achieves 3.86/5, and Takeaway achieves 4.08/5. The lower Delivery rating reflects inherent quality challenges (food temperature maintenance, packaging integrity, delivery time variability) that fall partially outside the restaurant's direct operational control. This insight motivates investment in insulated packaging, delivery partner quality standards, and delivery radius optimization.

E. Time-Based Analysis

Time-based analysis examines demand patterns across multiple temporal granularities: hourly, daily, weekly, monthly, and quarterly. Quarterly trend analysis using year-over-year comparison reveals consistent Q4 revenue strength (December quarter contributing 28.4% of annual revenue vs. a 25% equal-share baseline), confirming the holiday season demand premium. Q1 (January quarter) represents the weakest performance period at 21.3% of annual revenue, motivating January-specific promotional strategies.

Weekly periodicity analysis confirms that weekends (Friday-Sunday) generate 54.7% of total weekly revenue despite representing only 3/7 (42.9%) of days, indicating a 28% weekend revenue premium. Saturday emerges as the single

highest-revenue day of the week across all categories. This weekend concentration has significant staffing and inventory management implications, requiring approximately 40% greater kitchen and service capacity on weekends compared to weekday baseline levels.

Monthly heatmap analysis using a calendar heat map visualization reveals that the first week of each month exhibits slightly higher order volumes (approximately 7% above monthly average), potentially correlated with salary disbursement timing for many customers. This weekly-within-month pattern, while modest, suggests that time-limited promotional offers launched on the 1st of each month could capture elevated purchase intent more effectively.

XI. RESULTS AND DISCUSSION

A. Key Findings

The comprehensive analytical framework implemented through the Power BI dashboard has surfaced a series of interconnected findings that collectively characterize the restaurant operation's performance landscape. The primary finding of this research is the strong positive correlation (Pearson $r = 0.67$, $p < 0.01$) between a product category's average Sentiment Score and its monthly revenue growth rate, confirming that customer satisfaction is not merely a lagging indicator of quality but a leading predictor of commercial performance. Categories maintaining Positive Sentiment rates above 65% exhibited average monthly revenue growth of 4.2%, while those below 50% showed average monthly revenue contraction of 1.8%.

The second significant finding concerns order type channel evolution. Delivery channel order volumes grew at a 23.7% compound annual rate over the study period, compared to 4.2% for Dine-In and 11.3% for Takeaway. Despite the lower Average Rating associated with Delivery orders, revenue growth in this channel outpaced the others substantially, suggesting that the convenience-value proposition of delivery is sufficiently compelling to sustain demand growth even in the presence of quality trade-offs. However, the 0.35-point rating differential between Dine-In and Delivery represents a reputational risk that warrants proactive management.

The third finding relates to product portfolio composition. Analysis of the 45-product portfolio identifies a core group of 9 products (20%) responsible for 78% of total revenue — confirming the Pareto distribution hypothesis. Of these nine products, seven belong to the Main Course and Beverage categories. The remaining 36 products contribute only 22% of revenue while consuming proportional kitchen capacity, suggesting opportunities for menu rationalization and focused investment in the high-performing core product group.

B. Business Insights

Several high-value business insights emerge from the cross-dimensional analysis enabled by the Power BI framework. First, the identification of December and February as revenue peak months enables proactive capacity and procurement planning: kitchen inventory for peak-demand ingredients should be secured four to six weeks in advance to mitigate supply chain risks during periods of industry-wide elevated demand. Staffing augmentation plans should be formalized 30 days before peak months to allow adequate recruitment and training lead time.

Second, the correlation between customer loyalty tier and order frequency — with Platinum customers ordering 4.9x more frequently than Bronze customers — quantifies the revenue value of loyalty tier progression. Investing in Bronze-to-Silver and Silver-to-Gold tier advancement programs through targeted incentives (double-points events, tier-specific promotional offers) could yield disproportionate revenue returns given the multiplicative order frequency premium at higher tiers.

Third, the Fast Food category's above-average Negative sentiment rate (22.7% vs. 14.1% overall) combined with its 11.2% share of total revenue positions it as a high-risk, moderate-contribution category. A quality improvement initiative targeting the specific fast food items generating the most negative reviews — identifiable through the Product Insight Dashboard's product-level sentiment drill-down — could yield both direct revenue protection and brand perception improvement benefits.

C. Recommendations

Based on the analytical findings presented, the following strategic and operational recommendations are advanced for restaurant management consideration. Recommendation 1: Implement a Delivery Experience Enhancement Program targeting the 0.35-point Delivery-Dine-In rating gap. Specific initiatives should include packaging investment in insulated, leak-proof containers; delivery radius restriction to 5 km to reduce transit time; and implementation of a post-delivery satisfaction survey with automated service recovery messaging for ratings below 3.

Recommendation 2: Initiate a Menu Rationalization Review using the Product Insight Dashboard's revenue-per-product analysis to identify the lowest-performing 20% of products (by a composite score of revenue contribution, rating, and

sentiment) for potential menu removal or reinvention. Resources freed from supporting low-performing items should be redirected to premium development of high-performing core products.

Recommendation 3: Develop a Seasonal Revenue Optimization Calendar aligning promotional campaigns, limited-time menu offerings, and marketing spend concentration with the identified monthly revenue cyclical profile. January and June promotional investments should prioritize new customer acquisition and trial incentives, while December and February resources should focus on capacity management and experience quality preservation under high-volume conditions.

XII. CONCLUSION

This research has presented and demonstrated a comprehensive Power BI-based analytics framework for restaurant food trend analysis, customer sentiment measurement, and product insight generation. The framework successfully integrates multi-source restaurant datasets into a coherent relational data model, implements a structured ETL pipeline for data quality assurance, and delivers an interactive three-page dashboard providing actionable intelligence across food trends, customer experience, and product performance dimensions.

The analytical findings confirm several strategically significant relationships: the strong correlation between customer sentiment and revenue growth, the accelerating market share of delivery order channels despite their quality disadvantage, and the pronounced Pareto concentration of revenue within a small core product set. These findings generate concrete, evidence-based recommendations for delivery experience enhancement, menu rationalization, and seasonal promotional planning that are immediately actionable by restaurant management teams.

From a methodological perspective, this study demonstrates that Microsoft Power BI provides a sufficiently powerful analytical environment to support sophisticated multi-dimensional restaurant analytics without requiring dedicated data science infrastructure. The accessibility of Power BI's graphical ETL tools, relational modeling interface, and DAX calculation engine enables technically literate restaurant operators — or graduating computer science students applying their skills to real business problems — to construct and maintain this analytical framework independently.

The research contributes a replicable analytics template applicable to food service businesses of varying scale and format. The dataset structure, preprocessing methodology, star schema design, DAX measure library, and dashboard layout documented in this paper collectively constitute a transferable intellectual framework that can be adapted to diverse restaurant contexts by substituting the specific dataset while preserving the analytical architecture.

XIII. FUTURE WORK

A. AI and Machine Learning Integration

The most significant opportunity for enhancing the analytical framework described in this paper lies in the integration of machine learning-based predictive models. The current framework is fundamentally descriptive and diagnostic — it characterizes historical performance patterns and identifies their structural correlates. The natural evolution toward prescriptive analytics requires the development of predictive models capable of forecasting future demand, customer satisfaction trajectories, and product performance outcomes.

Microsoft Azure Machine Learning, which integrates directly with Power BI through the Azure ML connector, enables the deployment of trained predictive models as callable functions within Power Query and DAX. A demand forecasting model trained on the historical sales dataset using algorithms such as Facebook Prophet for time-series seasonality modeling or gradient-boosted regression trees for multi-feature demand prediction could be integrated into the Food Trends Dashboard to display forward-looking demand projections alongside historical actuals.

Natural Language Processing enhancements represent a particularly high-value AI integration opportunity. The current framework relies on pre-processed sentiment labels. Integration of a live sentiment scoring model — leveraging Azure Cognitive Services Text Analytics API — would enable real-time sentiment processing of new customer reviews as they are submitted, eliminating the batch processing lag and enabling near-real-time customer experience monitoring.

B. Real-Time Streaming Dashboards

The current dashboard architecture operates on batch-refreshed data, with reports updated on scheduled refresh cycles (hourly or daily, depending on Power BI licensing tier). For operational decision-making — particularly in high-volume restaurant environments where immediate visibility into current-hour sales performance and kitchen queue status is operationally critical — a real-time streaming dashboard architecture is required.

Power BI Streaming Datasets, enabled through the Power BI REST API, support the continuous ingestion of streaming event data into special dataset types that bypass the standard import/DirectQuery architecture in favor of push-based real-time data feeds. Integration with restaurant POS systems capable of emitting order-creation events through webhook APIs would enable real-time order volume, revenue, and average wait time monitoring on operational dashboards visible in the kitchen, at the management terminal, and on mobile devices.

C. Mobile Optimization and Accessibility

Future development should prioritize mobile-optimized dashboard layouts leveraging Power BI's dedicated phone layout designer, enabling restaurant managers to access key KPIs and trend visualizations from smartphone interfaces during service operations. Integration with Power BI's natural language Q&A feature would enable voice or text-based queries such as 'What was today's top-selling product?' or 'How does today's revenue compare to last Friday?' — reducing the barrier for non-technical users to extract ad-hoc insights without navigating complex filter interactions.

XIV. REFERENCES

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XV. APPENDIX

A. Sample DAX Measures Reference

The following DAX measures constitute the core calculation layer of the Power BI dashboard. These measures should be organized within a dedicated Measures table in the Power BI data model.

Measure Name	DAX Formula	Purpose
Total Revenue	SUMX(Orders_Fact, Orders_Fact[Qty] * Orders_Fact[Unit_Price] * (1 - Orders_Fact[Discount_Rate]))	Net revenue after discounts
Total Sales	COUNTROWS(Orders_Fact)	Transaction volume count
Avg Rating	AVERAGE(Reviews_Fact[Rating])	Mean customer satisfaction score
MoM Growth %	DIVIDE([Total Revenue] - CALCULATE([Total Revenue], DATEADD(Date_Dim[Date],-1,MONTH)), CALCULATE([Total Revenue], DATEADD(Date_Dim[Date],-1,MONTH)))	Month-over-month revenue change
Positive Sentiment %	DIVIDE(CALCULATE(COUNTROWS(Reviews_Fact), Reviews_Fact[Sentiment_Label]="Positive"), COUNTROWS(Reviews_Fact))	Proportion of positive reviews
Top Product	TOPN(1, VALUES(Products_Dim[Product_Name]), [Total Revenue], DESC)	Highest revenue product
YoY Growth %	DIVIDE([Total Revenue] - CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Date_Dim[Date])), CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Date_Dim[Date])))	Year-over-year revenue change

Table III. Core DAX Measures Reference — Formulas and Business Purpose

B. Data Model Field Glossary

Field Name	Table	Data Type	Description
Order_ID	Orders_Fact	Text	Unique alphanumeric order identifier
Order_Date	Orders_Fact	DateTime	Timestamp of order placement
Order_Type	Orders_Fact	Text	Channel: Dine-In, Takeaway, Delivery
Total_Amount	Orders_Fact	Decimal	Gross order value before discounts (INR)
Sentiment_Score	Reviews_Fact	Decimal	Continuous sentiment polarity (-1 to +1)
Sentiment_Label	Reviews_Fact	Text	Categorical: Positive, Neutral, Negative
Rating	Reviews_Fact	Integer	Customer star rating (1-5 scale)
Loyalty_Tier	Customers_Dim	Text	Bronze, Silver, Gold, Platinum
Category	Products_Dim	Text	Food category grouping (8 categories)
Month_Number	Date_Dim	Integer	Numeric month identifier (1-12)

Table IV. Data Field Glossary — Key Fields Across All Dataset Tables

C. Power BI Dashboard Development Checklist

The following checklist provides a structured implementation guide for replicating the dashboard described in this research paper in a new Power BI project environment.

Step 1 — Data Preparation: Ensure all five data tables are available as CSV or Excel files. Verify that Order_Date fields are in a consistent datetime format. Confirm that Product_ID and Customer_ID foreign keys are consistent across fact and dimension tables.

Step 2 — Power Query ETL: Load all five tables through 'Get Data > Text/CSV'. Apply data type conversions: Order_Date to DateTime, Rating to Integer, Total_Amount to Decimal. Add derived columns: Month_Year (text), Hour_of_Day (integer), Revenue_Band (conditional text column). Replace null values per the strategy documented in Section VI.C.

Step 3 — Data Modeling: Navigate to Model view. Establish relationships between fact tables and dimension tables as specified in Section VII.C. Mark Date_Dim as Date Table using the Date_Key field. Verify all relationship cardinalities and cross-filter directions.

Step 4 — Measures: Create a blank Measures table. Implement all DAX measures from Appendix A in the Measures table. Test each measure against expected values using a simple matrix visual before proceeding to dashboard construction.

Step 5 — Dashboard Construction: Create three report pages named 'Food Trends', 'Customer Sentiment', and 'Product Insights'. Add global slicers for Date Range, Year, and Category to all pages using the Sync Slicers feature. Construct the KPI header strip with four KPI cards on each page. Build page-specific visualizations as documented in Section VIII.

Step 6 — Formatting and Publication: Apply consistent color theme (Primary: #1a56db, Background: #f9fafb, Text: #374151). Enable drill-through between dashboard pages. Publish to Power BI Service. Configure scheduled data refresh if connecting to live data sources.

— End of Paper —